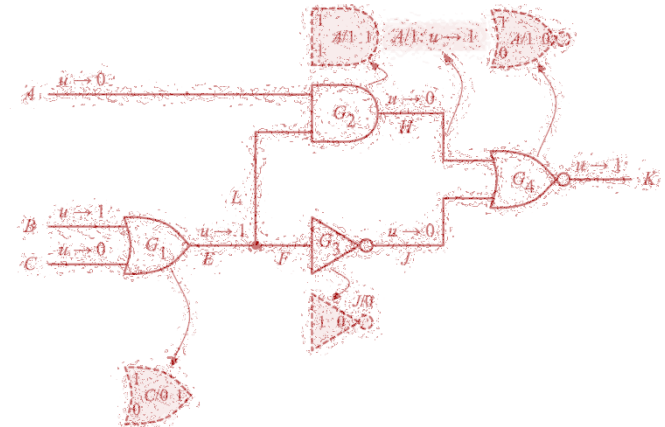


Fault Simulation

- Introduction
- Fault simulation techniques
- Alternatives to fault simulation
- Issues of fault simulation
- Conclusion

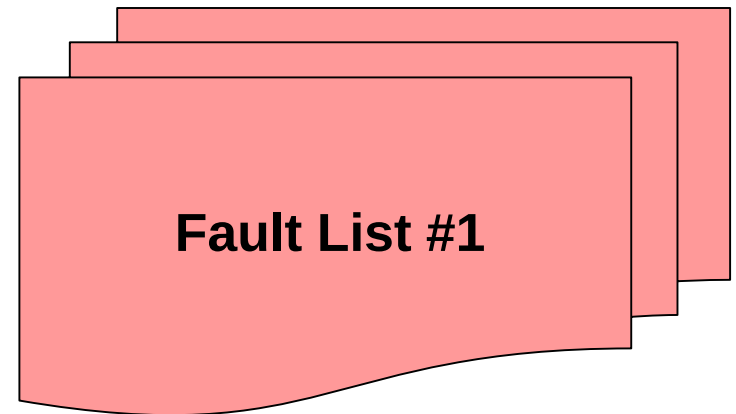


Issues of Fault Simulations

- Long CPU time
 - ◆ Distributed computing, or Fault sampling
- Large memory requirement
 - ◆ Partition faults into multiple simulation passes
- Potentially detected faults
- Compatibility with logic simulation

Speed and Memory Solutions

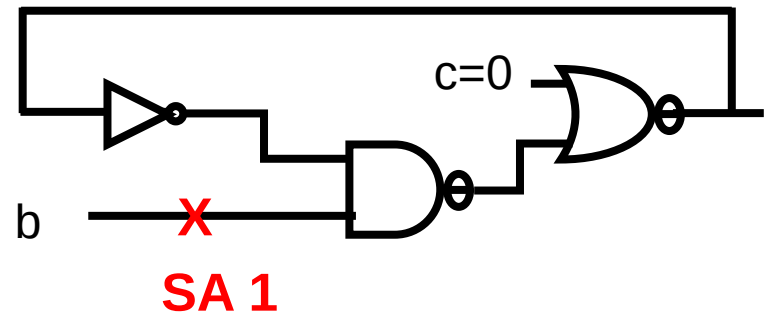
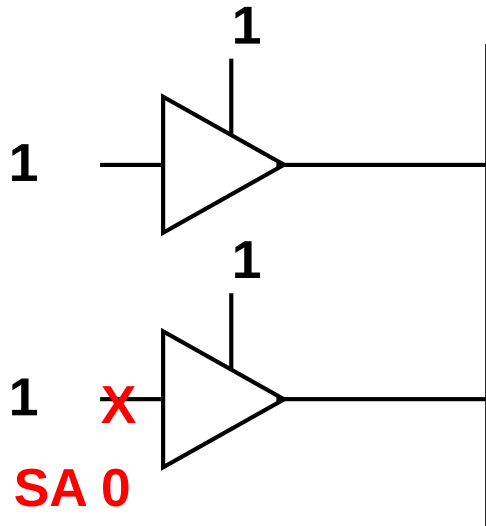
- 1. **Multiple-pass fault simulation**
 - ♦ Run only small portion of faults per pass
- 2. **Distributed fault simulation**
 - ♦ Distribute faults to more than one computers
- 3. **Emulation**
 - ♦ Use hardware emulator, like FPGA



- *A **simulation pass** is a single simulation run
 - ♦ from beginning to end of test patterns

Potentially Detected Faults

- **DEF**: faults that may or may not be detected in practice
 - ♦ Detection cannot be determined by fault simulation
- Possible reasons for potentially detected faults
 - ♦ Bus contention, Oscillation, High impedance, Unknown
- Examples:



Fault detected?

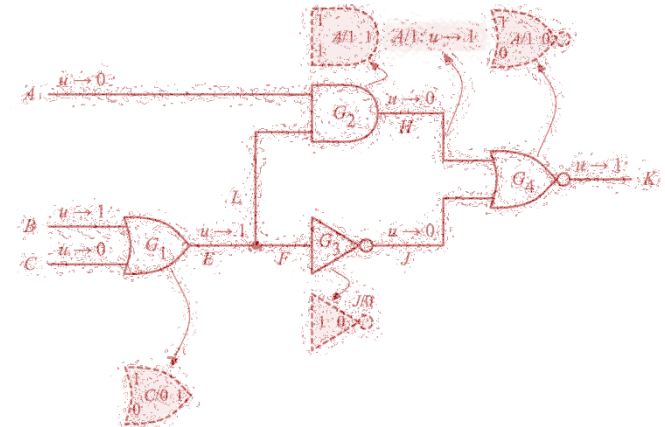
- Different tools have different ways to calculate FC
 - ♦ Please see tool manual for details

Compatibility with Logic Simulation

- To speed up fault simulation, many tools requires circuits to be represented in a library models pre-defined by tools
- Functional verification logic simulations often involve mixed-level codes, which make such modeling very difficult
 - ◆ Circuit delay
 - ◆ RTL behavior description
 - ◆ **User Defined Primitives (UDP)**
 - * DFF, MUX

Fault Simulation

- Introduction
- Fault simulation techniques
- Comparison of fault simulation
- Alternatives to fault simulation
- Issues of fault simulation
- Conclusion



Quiz: Comparison of Techniques

Items	Serial	PPSFP	Deductive	Concurrent	Differential
unknown logic value	😊	😊	😞	😊	😊
Delay model	x				
Run time	x				
Sequential circuits	x				
Memory	x				

Conclusion

- Usage of fault simulation
 - ◆ ATPG, fault grading, diagnosis
- Techniques
 - ◆ Serial, parallel, PPSFP, deductive, concurrent, differential
- Most popular technique in industry now
 - ◆ PPSFP is simple for comb. ckt (or seq. ckt with full scan)
 - ◆ Concurrent and differential good for both comb./seq. ckt

Commercial Tools

- **Cadance**
 - ◆ **Verifault**
- **Mentor Graphic:**
 - ◆ **Fastscan, flextest**
- **Synopsys**
 - ◆ **Tetramax**
- **Syntest**
 - ◆ **Turboscan**
 - ◆ **TurboFault**

References

- [Abramovici 1984] M. Abramovici, P. R. Menon, and D. T. Miller, “Critical Path Tracing: An Alternative to Fault Simulation,” IEEE Design and Test of Computers, 1984.
- [Armstrong 1972] D. B. Armstrong, A deductive method for simulating faults in logic circuits, IEEE Trans. Comput., C-21(5), 464–471, 1972.
- [Butler 1974] T. T. Butler, T. G. Hallin, J. J. Kulzer, and K. W. Johnson, LAMP: Application to switching system development, Bell System Tech. J., 53, 1535–1555, 1974.
- [Cheng 1989] W. T. Cheng and M. L. Yu, Differential fault simulation: A fast method using minimal memory, in Proc. Des. Automat. Conf., June 1989, pp. 424–428.
- [Jain 1985] S. K. Jain and V. D. Agrawal, Statistical fault analysis, IEEE Des. Test Comput., 2(1), 38–44, 1985.
- [Seshu 1965] S. Sesuh and D. N. Freeman, On improved diagnosis program, IEEE Trans. Electron. Comput., EC-14(1), 76–79, 1965.
- [Ulrich 1974] E. G. Ulrich and T. Baker, Concurrent simulation of nearly identical digital networks, IEEE Trans. Comput., 7(4), 39–44, 1974.
- [Waicukauski 1985] J. A. Waicukauski, E. B. Eichelberger, D. O. Forlenza, E. Lindbloom, and T. McCarthy, Fault simulation for structured VLSI, Proc. VLSI Syst. Des., 6(12), 20–32, 1985.